

# normality: A Rust crate for univariate and multivariate normality testing

Jezzu Morrisen C. Quimosing <sup>1</sup>

<sup>1</sup> School of Statistics, University of the Philippines Diliman 

DOI: [10.xxxxxx/draft](https://doi.org/10.xxxxxx/draft)

## Software

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Editor: 

Submitted: 19 April 2026

Published: unpublished

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## Summary

The assumption of normality is a common prerequisite for numerous parametric statistical tests, from  $t$ -tests to complex linear regression models. Validating this assumption is of importance. As discussed by Ferr (2025), real-world data often violates this assumption, exhibiting skewness, kurtosis, and outliers. Moreover, errors regarding the distribution of data are remarkably common in published scientific literature (Ghasemi & Zahediasl, 2012).

normality is a Rust library designed to address this by providing a suite of hypothesis tests for assessing both univariate and multivariate normality. By implementing widely used tests such as the Shapiro-Wilk, Anderson-Darling, Jarque-Bera, and Henze-Zirkler tests, normality provides researchers with the tools to validate the distribution of their data within the rapidly growing Rust ecosystem.

## Statement of need

Scientists and researchers alike are increasingly turning to Rust for scientific computing due to its C-like performance and guaranteed memory safety Bugden & Alahmar (2022). Recent studies in fields such as computational physics show that rewriting C++ simulations in Rust can yield performance increases while making parallelization safer and simpler to implement (Veytsman et al., 2024). As Rust expands heavily into big data engineering and data-intensive applications (?), there now is a need for statistical validation tools native to this ecosystem.

Currently, while researchers can visually assess distributions, further validation requires formal hypothesis testing. The misuse of these tests is prevalent; for instance, a review by Midway & White (2025) found that over 70% of ecology papers and 90% of biology papers incorrectly applied normality tests to raw data instead of regression model residuals. A well-documented Rust crate can help guide developers toward better and sound methodological practices.

Furthermore, literature indicates that no single normality test is universally the most powerful across all sample sizes and underlying distributions (Arnastauskaitė et al., 2021). Hence, a library like normality, which has multiple methods rather than a single approach, is needed for data science. It allows users to integrate the most appropriate checks into data pipelines and workflows where manual inspection is difficult.

## State of the field

The Rust ecosystem for data science and statistics has seen large growth over the years, establishing the groundwork for a complete scientific suite as documented in recent literature covering statistics with Rust (Nakamura, 2024). Several libraries currently exist:

- polars (Vink, 2020) provides a fast and multithreaded query engine for DataFrames;

- stats (Ma & @stats-dev, 2016) provides a set of statistical utilities, implementations of common distributions, and mathematical functions;
- polyfit (Carson, 2026) handles linear regression and curve fitting, also provides a collection of statistical tests to validate fit quality; and
- augurs (Sully, 2024) is a time series analysis and forecasting toolkit.

While these tools excel in their respective domains, there remains a gap in hypothesis testing for data distributions. normality fills this gap. It complements stats by providing the formal tests needed to see if data actually follows the theoretical distributions. Further, it also serves as a utility tool for libraries like polyfit and augurs, which can utilize normality's tests to validate model residuals—helping prevent the methodological errors discussed by Midway & White (2025).

Software design

The normality crate is designed to be modular, extensible, and mathematically accurate and rigorous. The implemented tests are the following:

| Name                            | Classification | Reference                     |
|---------------------------------|----------------|-------------------------------|
| Shapiro-Wilk                    | Univariate     | Shapiro & Wilk (1965)         |
| Lilliefors (Kolmogorov-Smirnov) | Univariate     | Lilliefors (1967)             |
| Anderson-Darling                | Univariate     | Anderson & Darling (1954)     |
| Jarque-Bera                     | Univariate     | Jarque & Bera (1987)          |
| Pearson Chi-squared             | Univariate     | Pearson (1900)                |
| D'Agostino's K-squared          | Univariate     | D'Agostino & Belanger (1990)  |
| Anscombe-Glynn                  | Univariate     | Anscombe & Glynn (1983)       |
| Energy Test                     | Univariate     | Móri et al. (2021)            |
| Henze-Zirkler                   | Multivariate   | N. Henze & Zirkler (1990)     |
| Henze-Wagner                    | Multivariate   | Norbert Henze & Wagner (1997) |
| Mardia                          | Multivariate   | Mardia (1970)                 |
| Pudelko                         | Multivariate   | Pudelko (2005)                |
| Székely-Rizzo                   | Multivariate   | Székely & Rizzo (2005)        |
| Royston                         | Multivariate   | Royston (1992)                |

Research impact statement

The crate's near-term significance is underpinned by its approach to validation and reproducibility. A feature normality's development cycle is its exhaustive cross-language testing framework. Software, particularly statistical software, requires absolute mathematical precision. To guarantee this, the crate utilizes an automated integration test that directly interfaces with the R statistical computing environment.

For every implemented test, the framework generates datasets that follow  $\mathcal{N}(0, 1)$  (normal) and  $U(0, 1)$  (uniform) across 28 distinct sample sizes (10-5000 observations). These datasets are simultaneously evaluated in Rust and by established packages in R. The test suite strictly enforces floating-point equivalence on test statistics and  $p$ -values in order to mathematically prove validity.

To ensure reproducibility, the crate is also tested across all major operating systems (Ubuntu, macOS, and Windows) across multiple Rust toolchains including an enforced Minimum Supported Rust Version (MSRV) of 1.87.0. Furthermore, to address the computational demands of large-scale data, the crate implements optional data-level parallelism, allowing researchers to parallelize sorting and complex statistical calculations with zero code modification.

This combination of mathematically verified accuracy, cross-platform stability, and scalable performance is what establishes normality as a credible utility for researchers building data-intensive applications and pipelines in Rust.

## AI usage disclosure

Google's Gemini was used by the author solely in order to check the grammatical accuracy and structure of this paper. No generative AI tools were used in the development of the library. The author confirms that human authorship was not superseded by checking and manually proofreading the above text.

## Acknowledgments

We acknowledge the R Core Team (R Core Team, 2025) and the authors of the R packages `nortest` (Gross & Ligges, 2006), `moments` (Komsta, 2005), and `MVN` (Korkmaz et al., 2014), from which the algorithms of this crate were inspired from. We also acknowledge the authors of the python package `scipy` (Virtanen et al., 2020) for their Cython implementation of the Shapiro-Wilk test.

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